# Campus Chakra



## 1. System objective

Campus Chakra is a campus resource-operations platform for the Prompt for the Planet challenge. It connects waste-event classification, resource telemetry forecasting, facility anomaly workflows and circular allocation views for an eight-building campus dataset.

### Edge layer

Smart-bin sensor payloads and gateway buffers continue locally during connectivity loss.

### Service layer

FastAPI validates uploads, runs deterministic models and returns auditable JSON.

### Experience layer

Dashboard turns telemetry into alerts, forecasts and allocation actions.

## 2. Architecture and control flow

1. **Ingest:** hourly streams, daily summaries, manual logs and waste events are uploaded or synchronized.
2. **Validate:** required columns, known building identifiers, timestamps and numeric measures are checked at the API boundary.
3. **Reconcile:** duplicate / delayed events are handled through deterministic identifiers and sequencing.
4. **Infer:** compact classifier, forecast bundle and anomaly logic run locally in the backend.
5. **Explain:** dashboard and optional copilot present evidence-led actions; deterministic fallback protects core flows.

## 3. Data assets

| Data family | Use in Campus Chakra |
|---|---|
| 8 building records + 120-day calendar | Campus context, occupancy/event alignment and building mapping. |
| 23,192 hourly telemetry rows | Forecasting, anomaly baselines and dashboard trends. |
| 415 manual logs | Alias normalization and operator-grounded incident context. |
| 6,314 train + 1,601 test waste events | Model development and required prediction-file generation. |

## 4. Data handling pipeline

| Stage | Implementation | Why it matters |
|---|---|---|
| Schema gate | API endpoint checks mandatory fields and type-convertible numeric values before modeling. | Prevents silent corruption from jury or gateway uploads. |
| Normalization | Maps building aliases and consistent units before aggregation. | Makes formal telemetry and operator-entered logs comparable. |
| Time integrity | Parses timestamps, sorts streams and uses idempotent gateway sequencing. | Supports delayed / retransmitted field data without double-counting. |
| Feature engineering | Sensor quality flags, temporal features and historic resource baselines are produced at inference. | Preserves repeatable, edge-compatible model behavior. |
| Serving | Models load from local cache; deterministic logic remains available when optional cloud services fail. | Maintains a usable evaluator demo under weak connectivity. |

## 5. Security and privacy approach

- Input validation reduces unsafe parsing and malformed-upload risk.
- Secrets are read from environment configuration; API keys are not embedded in client pages.
- The supplied challenge dataset is handled as a synthetic / competition dataset; production deployment would require institution-approved access controls, retention policy and consent review.
- Recommendations are operational decision support, with facility staff retaining approval authority.

## 6. Failure handling

> **Offline first:** classification, forecast and integrity paths do not depend on a live third-party LLM. The copilot can enrich explanations, but core predictions remain deterministic.

- Invalid uploads return readable HTTP validation errors.
- Missing or unavailable optional services return a known fallback status rather than blocking the dashboard.
- Gateway synchronization uses a high-resolution sequence to avoid collisions during rapid retries.

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## 7. Models and evaluator workflow

| Capability | Approach | Evaluator-facing output |
|---|---|---|
| Waste sorting | Compact random-forest classifier using physical sensor readings and quality indicators. | Predicted class plus confidence; test output in `waste_events_test_predictions.csv`. |
| Demand outlook | Short-horizon feature bundle for electricity and water demand. | 6h / 12h forecast values, direction and risk context. |
| Facility alerts | Rule-based and statistical signals joined with building and event context. | Prioritized alert cards with suggested next action. |
| Circular routing | Class-aware allocation logic maps material streams to recovery paths. | Waste allocation / recovery summary for operations teams. |

## 8. Validation performed

- Dashboard endpoint response check.
- Single-event classifier check and full 1,601-row waste test-file upload check.
- Hourly-stream forecast upload check and manual-log evaluation check.
- Gateway synchronization retry / duplicate behavior check.
- Prediction CSV verified to contain exactly `event_id,predicted_class` and 1,601 predictions.

## 9. Scalability and next steps

The service keeps core inference local and stateless per request, allowing buildings to be added primarily through configuration and telemetry onboarding. A production rollout would add authenticated users, encrypted durable storage, monitoring, calibration procedures and a scheduled retraining / model-governance process.

> **Evaluation boundary:** reported functionality is based on the supplied competition dataset and local smoke tests. It should be assessed as a working competition prototype, not as a claim of live campus deployment.